

Student Name: _____

2014

Year 12 Extension 1 Mathematics

Trial Examination

Teacher Setting Paper: Mrs Northam

Head of Department: Mrs Hill

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
(Black pen is preferred)
- Board approved calculator may be used
- Write your answers for Section I on the multiple answer sheet provided
- Write your answers to Section II on the paper provided. Start a new sheet for each question
- Write your student number only at the top of each page
- A table of standard integrals is provided at the back of this paper
- In Questions 11- 14, show relevant mathematical reasoning and/or calculations.

Total marks – 70

Section I – Multiple Choice

10 marks

Attempt Questions 1-10

Allow 15 minutes for this section

Section II – Extended Response

60 marks

Attempt questions 11 - 14

Allow 1 hours and 45 minutes for this section

*This examination paper does not necessarily reflect the content or format of the
Higher School Certificate Examination in this subject*

Section I

10 marks

Attempt Questions 1 – 10

Use the multiple-choice answer sheet for Questions 1 - 10.

1. What is the value of $\lim_{x \rightarrow 0} \frac{\sin 3x}{2x}$

(A) 0

(B) $\frac{2}{3}$

(C) 1

(D) $\frac{3}{2}$

2. The remainder obtained when $P(x) = x^3 - 3x^2 - 5x + 6$ is divided by $x - 3$ will be

(A) -34

(B) -9

(C) 6

(D) 21

3. How many distinct permutations of the letters of the word 'ATTAINS' are possible in a straight line when the word begins and ends with the letter T?

(A) 60

(B) 120

(C) 360

(D) 1260

4. What is the sixth term in the expansion of $(2x - 3y)^9$?

(A) ${}^9C_3 \times 2^6 \times (-3)^3 x^6 y^3$

(B) ${}^9C_4 \times 2^5 \times (-3)^4 x^5 y^4$

(C) ${}^9C_5 \times 2^4 \times (-3)^5 x^4 y^5$

(D) ${}^9C_6 \times 2^3 \times (-3)^6 x^3 y^6$

5. The derivative of $\tan^{-1} 2x$ is

(A) $\frac{2}{4+x^2}$

(B) $\frac{2}{1+4x^2}$

(C) $\frac{1}{4+x^2}$

(D) $\frac{1}{1+4x^2}$

6. Which of the following is the range of the function $y = 4 \cos^{-1} 3x$

(A) $-\frac{1}{3} \leq y \leq \frac{1}{3}$

(B) $-\frac{1}{4} \leq y \leq \frac{1}{4}$

(C) $0 \leq y \leq \frac{\pi}{4}$

(D) $0 \leq y \leq 4\pi$

7. Consider the function $f(x) = x^2 - 6x$. Which of the following gives the correct domain of $f(x)$ for which there exists an inverse function, $f^{-1}(x)$.

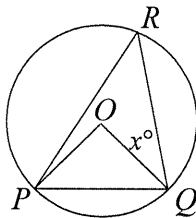
(A) All real x

(B) $0 \leq x \leq 6$

(C) $x \leq 3$ or $x \geq 3$

(D) $x \geq 0$

8. Consider the diagram below.



PQ is a chord of a circle, centre O , and R is a point on the major arc.

If $\angle OPR = 5^\circ$, $\angle OQP = 40^\circ$, then the value of x is:

- (A) 30
- (B) 35
- (C) 40
- (D) 45
9. A particle is moving in a straight line with $v^2 = 36 - 4x^2$ and undergoing simple harmonic motion. If the particle is initially at the origin, which of the following is the correct equation for its displacement in terms of t ?
- (A) $x = 2 \sin 3t$
- (B) $x = 3 \sin 2t$
- (C) $x = 2 \sin 9t$
- (D) $x = 3 \sin 4t$
10. Eden, Toby and four friends arrange themselves at random in a circle. What is the probability that Eden and Toby are *not* together?
- (A) $\frac{1}{120}$
- (B) $\frac{2}{5}$
- (C) $\frac{3}{5}$
- (D) $\frac{119}{120}$

END OF SECTION I

Section II

90 marks

Attempt Questions 11-14

Write your answers on the paper provided

Question 11 (15 marks) **Start a new page.**

- (a) Find the coordinates of the point P, which divides the interval from A(-1,8) to B(13,3) internally in the ratio 5:2. 2
- (b) Find to the nearest minute, the acute angles between the lines $y = -\frac{x}{3} + 4$ and $y = x + 1$ 2
- (c) Using the substitution $u = e^{2x}$ find the value of $\int \frac{e^{2x}}{1 + e^{4x}} dx$ 3
- (d) Solve $\frac{3}{2x - 4} > -2$ 3
- (e) Differentiate $e^x \tan^{-1} 3x$ 2
- (f) The polynomial equation $2x^3 - 3x^2 + 4x - 7 = 0$ has roots α, β and γ . 3
Find the exact value of $\frac{1}{\alpha\beta} + \frac{1}{\alpha\gamma} + \frac{1}{\beta\gamma}$.

Question 12 (15 marks) **Start a new page.**

(a) Consider the function $f(x) = 2 \cos^{-1} \frac{x}{3}$

(i) Evaluate $f(0)$. 1

(ii) State the domain and range of this function. 2

(ii) Sketch the graph of $f(x) = 2 \cos^{-1} \frac{x}{3}$ over the stated domain. 3

(b) Find $\int \sin^2 2x dx$ 2

(c) Find the value of the term independent of x in the expansion of $\left(x^2 - \frac{1}{x}\right)^{12}$ 3

(d) The chord PQ of the parabola $x^2 = 4y$ subtends a right angle at the origin O .

If the coordinates of P and Q are $(2p, p^2)$ and $(2q, q^2)$ respectively:

(i) Find the gradients of PO and QO . 1

(ii) Show that $pq = -4$. 1

(iii) Find the equation of the locus of the midpoint M of PQ . 2

Question 13 (15 marks) Start a new page.

(a) Find $\int \frac{1}{\sqrt{3-4x^2}} dx$ 2

- (b) At time t minutes, the temperature T° Celsius of an object is given by $T = 24 - 22e^{-kt}$ where k is a constant.

After 5 minutes the temperature of the object has risen from $2^\circ C$ to $13^\circ C$.

- (i) Find the exact value of k 2

- (ii) Find the temperature of the object after 10 minutes. 1

- (c) When a particle is x metres from the origin, its velocity, $v \text{ ms}^{-1}$, is given by 2

$$v = \sqrt{8 - 2x^2}.$$

Find the acceleration when the particle is 2 metres to the right of the origin.

- (d) A computer animation shows the sides of a cube increasing at a rate of 3mm/s. 3

Find the rate at which the volume V is increasing when the cube has a side length of 5mm.

- (e) Find the exact value of $\cos\left(\sin^{-1}\left(-\frac{1}{3}\right)\right)$ 2

- (f) Use the Principle of Mathematical Induction to prove that, for all positive integers, n , 3

$$\sum_{r=1}^n \frac{1}{(4r-3)(4r+1)} = \frac{n}{4n+1}$$

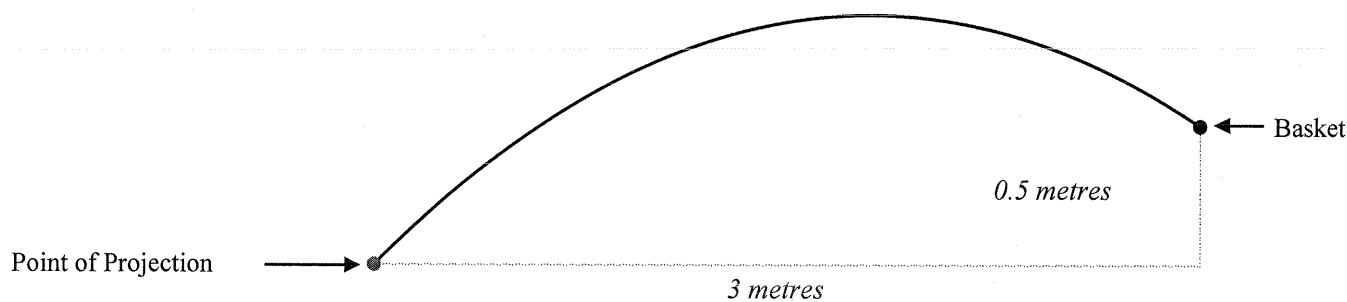
Question Fourteen (15 marks) **Start a new page.**

- (a) Greg is about to have a shot at goal in a game of basketball.

From the point where the ball leaves his hand, the distance to the top of the basket is 3 metres horizontally and 0.5m vertically. Greg shoots at the optimal angle of 45° .

You may assume the equations of motion are

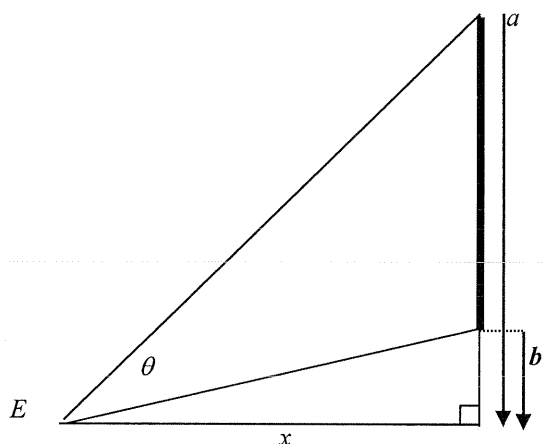
$$x = vt \cos 45^\circ \text{ and } y = vt \sin 45^\circ - 5t^2 \text{ (Do NOT prove this)}$$



- (i) Find the velocity of projection, v , required by Greg for the centre of the ball to land in the centre of the basket. 2
- (ii) Find the maximum vertical height above the basket that the ball reaches during Greg's shot. 2
- (iii) Find the speed of the ball on entry into the basket. 2

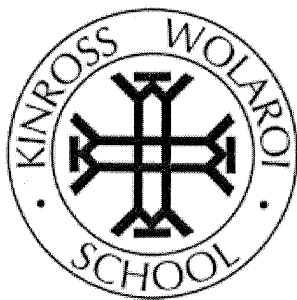
- (b) If $(1+x)^n = \sum_{r=0}^n {}^nC_r x^r$ prove $\sum_{r=1}^n r {}^nC_r = n \cdot 2^{n-1}$ 3

- (c) An observer's eye E looks up at a large painting on a vertical wall. The top of the painting is a metres above the level of E and the bottom of the painting is b metres above the level of E . θ is the angle subtended at the observer's eye by the top and bottom of the painting. E is x metres from the wall. The observer can move backwards and forwards changing x to find the position of best view when θ is a maximum.



- (i) Explain why $\theta = \tan^{-1} \frac{a}{x} - \tan^{-1} \frac{b}{x}$ 1
- (ii) Show that $\frac{d\theta}{dx} = \frac{(a-b)(ab-x^2)}{(a^2+x^2)(b^2+x^2)}$ 3
- (iii) If $a = 3b$, find the maximum possible value for θ . 2

END OF SECTION II



2014

Year 12 Extension 1 Mathematics
Yearly Examination

MULTIPLE CHOICE ANSWER SHEET

For multiple choice questions, choose the best answer A, B, C or D and fill in the correct circle.

1. ☐ A ☐ B ☐ C ☐ D

2. ☐ A ☐ B ☐ C ☐ D

3. ☐ A ☐ B ☐ C ☐ D

4. ☐ A ☐ B ☐ C ☐ D

5. ☐ A ☐ B ☐ C ☐ D

6. ☐ A ☐ B ☐ C ☐ D

7. ☐ A ☐ B ☐ C ☐ D

8. ☐ A ☐ B ☐ C ☐ D

9. ☐ A ☐ B ☐ C ☐ D

10. ☐ A ☐ B ☐ C ☐ D

Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax \, dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax \, dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax \, dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax \, dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x$, $x > 0$

2014 Extension 1 Trial Paper Solutions.

$$1. \lim_{x \rightarrow 0} \frac{\sin 3x}{2x}$$

$$= \frac{3}{2} \lim_{x \rightarrow 0} \frac{\sin 3x}{3x}$$

$$= \frac{3}{2} \quad (D)$$

$$2. P(3) = 3^3 - 3 \times 3^2 - 5 \times 3 + 6$$

$$= -9 \quad (B)$$

$$3. \frac{5!}{2!} \quad (A)$$

$$= 60 \quad (A)$$

$$4. {}^9C_5 \quad (C)$$

$$5. \frac{1}{1+(2x)^2} \times 2$$

$$= \frac{2}{1+4x^2} \quad (B)$$

$$6. 0 \leq \frac{y}{4} \leq \pi$$

$$0 \leq y \leq 4\pi \quad (D)$$

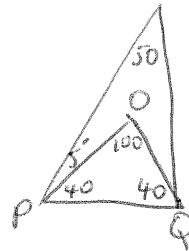
$$7. f(x) = x^2 - 6x$$



$$x \leq 3 \quad x \geq 3$$

horizontal straight line test

8.



$$45 + 50 + 40 + x = 180$$

$$x = 45$$

(D)

$$9. v^2 = 36 - 4x^2 \quad t=0 \quad x=0$$

$$v^2 = 4(9 - x^2) \quad v^2 = n^2(a^2 - x^2)$$

$$\therefore n=2 \quad a=3$$

$$\therefore x = a \sin nt$$

$$x = 3 \sin 2t \quad (B)$$

10. All ways in a circle $5! = 120$

Eden + Toby together

ET

□

□

$$4! \times 2$$

□ □

$\therefore$ Eden + Toby not together

$$120 - 4! \times 2 = 72$$

$$\frac{72}{120} = \frac{3}{5} \quad (C)$$

1. D

6. D

2. B

7. C

3. A

8. D

4. C

9. B

5. B

10. C

Question 11

(a) $(-1, 8)$ $(13, 2)$
 $5:2$

$$\left(\frac{5 \times 13 + 2 \times (-1)}{5+2}, \frac{5 \times 2 + 2 \times 8}{5+2} \right)$$

$$\left(9, \frac{31}{7} \right) \text{ or } \left(9, 4\frac{3}{7} \right)$$

(b) $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

$$m_2 = -\frac{1}{3} \quad m_1 = 1$$

$$\tan \theta = \left| \frac{1 - (-\frac{1}{3})}{1 + (-\frac{1}{3}) \times 1} \right|$$

$$\tan \theta = \left| \frac{\frac{4}{3}}{-\frac{2}{3}} \right|$$

$$\tan \theta = 2$$

~~... ..~~
~~... ..~~
~~... ..~~

(c) $u = e^{2x} = 63^\circ 26' 5.82''$

$$\frac{du}{dx} = 2e^{2x}$$

$$\frac{1}{2} \int \frac{2e^{2x}}{1+e^{4x}} dx$$

$$\frac{1}{2} \int \frac{1}{1+u^2} du$$

$$= \frac{1}{2} \tan^{-1} u + C$$

$$= \frac{1}{2} \tan^{-1} e^{2x} + C$$

(d) $\frac{3}{2x-4} > -2$

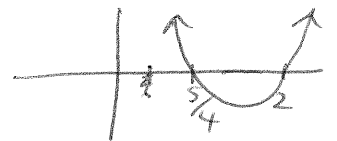
$$(2x-4)^2 \times \frac{3}{2x-4} > -2(2x-4)^2$$

$$3(2x-4) > -2(2x-4)^2$$

$$3(2x-4) + 2(2x-4)^2 > 0$$

$$(2x-4)[3 + 2(2x-4)] > 0$$

$$2(x-2)[4x-5] > 0$$



$$x < \frac{5}{4} \quad x > 2$$

(e) Product rule.

$$e^x \tan^{-1} 3x + e^x \times \frac{1}{1+(3x)^2} \times 3$$

$$e^x \tan^{-1} 3x + \frac{3e^x}{1+9x^2}$$

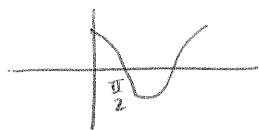
(f) $\frac{1}{\alpha\beta} + \frac{1}{\alpha\gamma} + \frac{1}{\beta\gamma}$

$$= \frac{\gamma + \beta + \alpha}{\alpha\beta\gamma}$$

$$= \frac{-\frac{3}{2}}{-\frac{7}{2}} = \frac{3}{7}$$

Question 12

(a)(i) $f(0) = 2\cos^{-1} 0$

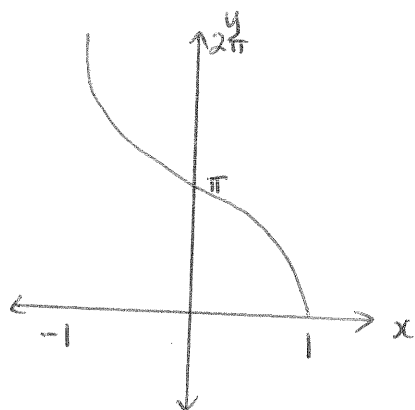


$$f(0) = 2 \times \frac{\pi}{2} = \pi$$

(ii) Domain: $-1 \leq \frac{x}{3} \leq 1$
 $-3 \leq x \leq 3$

Range $0 \leq \frac{y}{2} \leq \pi$
 $0 \leq y \leq 2\pi$

(iii)



(b) $\cos 4x = 1 - 2\sin^2 2x$

$$2\sin^2 2x = 1 - \cos 4x$$

$$\sin^2 2x = \frac{1}{2}(1 - \cos 4x)$$

$$\therefore \int \sin^2 2x = \frac{1}{2} \int (1 - \cos 4x) dx$$

$$\frac{1}{2} \left[x - \frac{\sin 4x}{4} \right] + C$$

$$= \frac{1}{2}x - \frac{\sin 4x}{8} + C$$

(c) ${}^{12}C_k (x^2)^{12-k} \left(-\frac{1}{x}\right)^k$

$$(x^2)^{12-k} \times (x^{-1})^k$$

$$24 - 2k - k = 0$$

$$24 - 3k = 0$$

$$k = 8$$

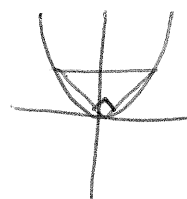
$${}^{12}C_8 (-1)^8$$

$$= {}^{12}C_8 = 495$$

(d)(i) $m_{p0} = \frac{p^2 - 0}{2p - 0}$

$$= \frac{p}{2}$$

$$m_{q0} = \frac{q^2 - 0}{2q} = \frac{q}{2}$$



(ii) If the chord subtends a right angle at the origin

$$m_1 \times m_2 = -1$$

$$\frac{p}{2} \times \frac{q}{2} = -1$$

$$pq = -4$$

(iii) $M\left(\frac{2p+2q}{2}, \frac{p^2+q^2}{2}\right)$

$$\left(p+q, \frac{p^2+q^2}{2}\right)$$

$$x = p+q \quad y = \frac{(p+q)^2}{2} - 2pq$$

$$\therefore y = \frac{x^2}{2} - 2x - 4$$

$$y = \frac{x^2}{2} + 8$$

$$2y = x^2 + 16$$

$$x^2 = 2y - 16$$

Question 13

$$(f) \sum_{r=1}^n \frac{1}{(4r-3)(4r+1)} = \frac{n}{4n+1}$$

Step 1:

Prove true for $n=1$.

$$\text{LHS } \frac{1}{(4 \times 1 - 3)(4 \times 1 + 1)} = \frac{1}{1 \times 5} = \frac{1}{5}$$

$$\text{RHS } \frac{1}{4 \times 1 + 1} = \frac{1}{5}$$

$$\text{LHS} = \text{RHS}$$

$\therefore$ true for $n=1$

Step 2: Assume true for $n=k$

$$\frac{1}{(4 \times 1 - 3)(4 \times 1 + 1)} + \frac{1}{(4 \times 2 - 3)(4 \times 2 + 1)} + \dots + \frac{1}{(4k-3)(4k+1)} = \frac{k}{4k+1}$$

Step 3: Prove true for $n=k+1$

$$\frac{1}{(4 \times 1 - 3)(4 \times 1 + 1)} + \dots + \frac{1}{(4k-3)(4k+1)} + \frac{1}{(4(k+1)-3)(4(k+1)+1)} = \frac{k+1}{4(k+1)+1}$$

$$\frac{1}{(4 \times 1 - 3)(4 \times 1 + 1)} + \dots + \frac{1}{(4k-3)(4k+1)} + \frac{1}{(4k+1)(4k+5)} = \frac{k+1}{4k+5}$$

$$\text{LHS } \frac{1}{(4 \times 1 - 3)(4 \times 1 + 1)} + \dots + \frac{1}{(4k-3)(4k+1)} + \frac{1}{(4k+1)(4k+5)}$$

$$= \frac{k}{4k+1} + \frac{1}{(4k+1)(4k+5)}$$

$$= \frac{k(4k+5) + 1}{(4k+1)(4k+5)}$$

$$= \frac{4k^2 + 5k + 1}{(4k+1)(4k+5)}$$

$$= \frac{(4k+1)(k+1)}{(4k+1)(4k+5)}$$

$$= \frac{(k+1)}{(4k+5)} = \text{RHS} \therefore$$

Since $n=1$ is true then
 $n=2$ is true etc

$\therefore$ Proved by
Mathematical
Induction.

Question 13

$$(a) \int \frac{1}{\sqrt{3-4x^2}} dx$$

$$\int \frac{1}{\sqrt{4(\frac{3}{4}-x^2)}} dx$$

$$\frac{1}{2} \int \frac{1}{\sqrt{\frac{3}{4}-x^2}} dx$$

$$\frac{1}{2} \sin^{-1} \frac{x}{\frac{\sqrt{3}}{2}} + C$$

$$\frac{1}{2} \sin^{-1} \frac{2x}{\sqrt{3}} + C$$

$$(b) (i) T = 24 - 22e^{-kt}$$

$$t=5 \quad 13 = 24 - 22e^{-5k}$$

$$T=13^\circ\text{C} \quad -11 = -22e^{-5k}$$

$$e^{-5k} = \frac{1}{2}$$

$$\ln e^{-5k} = \ln \frac{1}{2}$$

$$-5k = \ln \frac{1}{2}$$

$$k = \frac{\ln \frac{1}{2}}{-5}$$

$$k = \frac{\ln 2^{-1}}{-5}$$

$$k = \frac{-\ln 2}{-5}$$

$$k = \frac{\ln 2}{5}$$

$$(ii) T = 24 - 22e^{\frac{-\ln 2}{5} \times 10}$$

$$T = 18.5^\circ\text{C}$$

$$(c) \quad V = \sqrt{8-2x^2} \quad v > 0$$

$$\ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

$$\ddot{x} = \frac{d}{dx} \left(\frac{1}{2} (8-2x^2) \right)$$

$$\ddot{x} = \frac{d}{dx} (4-x^2)$$

$$\ddot{x} = -2x$$

$$\text{When } x=2$$

$$\ddot{x} = -4 \text{ m/s}^2$$

(d) Let the side of a cube be s

$$\frac{ds}{dt} = 3 \quad \frac{dV}{dt} = ?$$

$$\frac{dV}{dt} = \frac{ds}{dt} \times \frac{dV}{ds} \quad \checkmark$$

$$V = s^3$$

$$\frac{dV}{ds} = 3s^2 \quad \checkmark \text{ When } s=5$$

$$= 3 \times 5^2$$

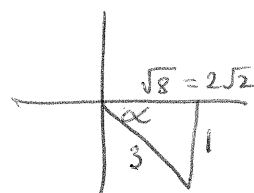
$$= 75$$

$$\therefore \frac{dV}{dt} = 3 \times 75 = 225 \text{ mm}^3/\text{s} \quad \checkmark$$

$$(e) \cos \left(\sin^{-1} \left(-\frac{1}{3} \right) \right)$$

$$\cos \left(-\sin^{-1} \frac{1}{3} \right)$$

$$= \frac{2\sqrt{2}}{3}$$



Question 14

(a) (i) $x = vt \cos 45$
 $y = vt \sin 45 - 5t^2$

$x = 3$ $y = 0.5$

$x = vt \cos 45$

$3 = vt \times \frac{1}{\sqrt{2}}$

$3\sqrt{2} = vt$

$t = \frac{3\sqrt{2}}{v}$

$y = vt \sin 45 - 5t^2$

$0.5 = v \left(\frac{3\sqrt{2}}{v} \right) \times \frac{1}{\sqrt{2}} - 5 \left(\frac{3\sqrt{2}}{v} \right)^2$

$0.5 = 3 - 5 \times \frac{9 \times 2}{v^2}$

$-2.5 = -5 \times \frac{18}{v^2}$

$v^2 = \frac{-5 \times 18}{-2.5}$

$v^2 = 36$

$v = 6 \text{ m/s}$ since $v > 0$

(ii) Max height $\dot{y} = 0$

$y = vt \sin 45 - 5t^2$

$\dot{y} = v \sin 45 - 10t$

$0 = 6 \times \frac{1}{\sqrt{2}} - 10t$

$10t = \frac{6}{\sqrt{2}}$

$t = \frac{6}{10\sqrt{2}} = \frac{3}{5\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{3\sqrt{2}}{10} \text{ seconds.}$

(iii) continued.

$y = 6 \times \frac{3\sqrt{2}}{10} \times \frac{1}{\sqrt{2}} - 5 \left(\frac{3\sqrt{2}}{10} \right)^2$

$y = 0.9 \text{ m}$

Which is 0.4 m (40 cm) above the basket.

(iii) When $x = 3$

$t = \frac{3\sqrt{2}}{6} = \frac{\sqrt{2}}{2} \text{ seconds.}$

$\dot{x} = v \cos 45$

$\dot{y} = v \sin 45 - 10t$

$\dot{x} = 6 \times \frac{1}{\sqrt{2}}$

$\dot{y} = 6 \times \frac{1}{\sqrt{2}} - 10 \times \frac{\sqrt{2}}{2}$

$= \frac{6}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$

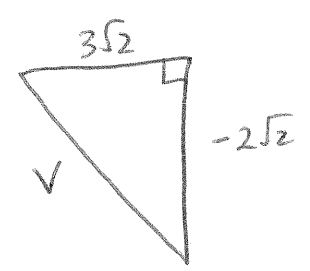
$\dot{y} = \frac{6}{\sqrt{2}} - 5\sqrt{2}$

$= 3\sqrt{2}$

$\dot{y} = \frac{6}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} - 5\sqrt{2}$

$\dot{y} = 3\sqrt{2} - 5\sqrt{2}$

$= -2\sqrt{2}$



$v^2 = (3\sqrt{2})^2 + (-2\sqrt{2})^2$

$v^2 = 18 + 8$

$v^2 = 26$

$v = \pm \sqrt{26} \text{ m/s}$

$\therefore \text{Speed} = \sqrt{26} \text{ m/s.}$

Question 14

$$(b) (1+x)^n = \sum_{r=0}^n {}^nC_r x^r$$

$$= {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \dots + {}^nC_n x^n$$

Prove $\sum_{r=1}^n r {}^nC_r = n 2^{n-1}$

$$\therefore {}^nC_1 + 2 {}^nC_2 + 3 {}^nC_3 + \dots + n {}^nC_n = n 2^{n-1}$$

$$(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \dots + {}^nC_n x^n$$

Differentiate:

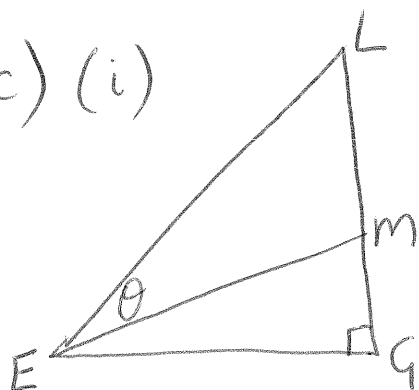
$$n(1+x)^{n-1} = {}^nC_1 + 2 {}^nC_2 x + 3 {}^nC_3 x^2 + \dots + n {}^nC_n x^{n-1}$$

Let $x=1$

$$n \times 2^{n-1} = {}^nC_1 + 2 {}^nC_2 + 3 {}^nC_3 + 4 {}^nC_4 + \dots + n {}^nC_n$$

$\therefore$ proved.

(c) (i)



$$\tan \angle LEG = \frac{a}{x} \quad \therefore \angle LEG = \tan^{-1} \frac{a}{x}$$

$$\tan \angle MEG = \frac{b}{x} \quad \angle MEG = \tan^{-1} \frac{b}{x}$$

$$\theta = \angle LEG - \angle MEG$$

$$= \tan^{-1} \frac{a}{x} - \tan^{-1} \frac{b}{x}$$

(ii) $\theta = \tan^{-1} \frac{a}{x} - \tan^{-1} \frac{b}{x}$

$$\frac{d\theta}{dx} = \frac{1}{1 + \left(\frac{a}{x}\right)^2} \times -ax^{-2} - \frac{1}{1 + \left(\frac{b}{x}\right)^2} \times -bx^{-2}$$

$$\frac{d\theta}{dx} = \frac{1}{1 + \left(\frac{a}{x}\right)^2} \times \frac{-a}{x^2} - \frac{1}{1 + \left(\frac{b}{x}\right)^2} \times \frac{-b}{x^2}$$

$$= \frac{1}{1 + \frac{a^2}{x^2}} \times \frac{-a}{x^2} - \frac{1}{1 + \left(\frac{b}{x}\right)^2} \times \frac{-b}{x^2}$$

$$= \frac{1}{\frac{x^2 + a^2}{x^2}} \times \frac{-a}{\cancel{x^2}} - \frac{1}{\frac{x^2 + b^2}{x^2}} \times \frac{-b}{\cancel{x^2}}$$

$$= \frac{-a}{x^2 + a^2} + \frac{b}{x^2 + b^2}$$

$$= \frac{-a(x^2 + b^2) + b(x^2 + a^2)}{(x^2 + a^2)(x^2 + b^2)}$$

$$= \frac{-ax^2 - ab^2 + bx^2 + a^2b}{(x^2 + a^2)(x^2 + b^2)}$$

$$= \frac{-x^2(a-b) + ab(-b+a)}{(x^2 + a^2)(x^2 + b^2)}$$

$$= \frac{(a-b)[-x^2 + ab]}{(x^2 + a^2)(x^2 + b^2)}$$

$$= \frac{(a-b)(ab - x^2)}{(x^2 + a^2)(x^2 + b^2)}$$